

Global Markets Comment: Distinguishing AI Risks—Size of the Pie vs. Share of the Pie

- We showed recently that as the value built into AI-related equities has grown, more optimistic assumptions about the ultimate economic gains for US companies from AI are needed to justify it. This likely makes the market more vulnerable to challenges to the more optimistic story. Although robust earnings may dominate valuation concerns for now, we think investors should focus on the potential challenges to the theme and their consequences for portfolios.
- In weighing those risks, a key distinction is whether a shock or risk shifts views on the likely *aggregate* value of AI-related innovation or on the *distribution* of that value (or both). Put simply, some worries are about the size of the pie and some are about how the pie is shared. We illustrate that this distinction is critical to understanding the impact of different challenges to the AI theme on markets and volatility.
- Only *aggregate* shocks are likely to have reliable, broad cross-asset impacts. For those with broad exposure to US equities, these are the shocks that matter most and macro hedges are likely to be available to protect against them. For those with more specific AI-related exposures, hedging is more challenging, since *distributional* shocks may also affect portfolios but have less reliable impacts on macro assets.
- For *aggregate* shocks to AI exposures, equity indices and rates so far appear to show clearer responses than FX and commodities. Except when higher rates are themselves the source of the risk, worries about the aggregate gains from AI are likely to push US yields lower. Aggregate shocks also have more scope to push equity index volatility higher than distributional shocks.

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Distinguishing AI Risks—Size of the Pie vs. Share of the Pie

Pressure on semiconductor stocks over the last month has revived questions about the potential risks to AI-related equities and ways to hedge against those risks.

A month ago, we looked again at what a macro approach to valuing the gains from AI says about the value that the market has assigned to AI relative to what the economy can deliver. That exercise showed that the value that had been built into AI-related equities has continued to grow and that more optimistic assumptions about the Present Discounted Value (PDV) of economic gains from AI that will accrue to companies were now needed to justify it. We also highlighted how the market has disproportionately rewarded the companies supplying the AI investment boom over

the last 6–9 months. So alongside the risks to the market view of the *aggregate* gains from AI, there might also be risks to the market’s view of the *distribution* of winners and losers.

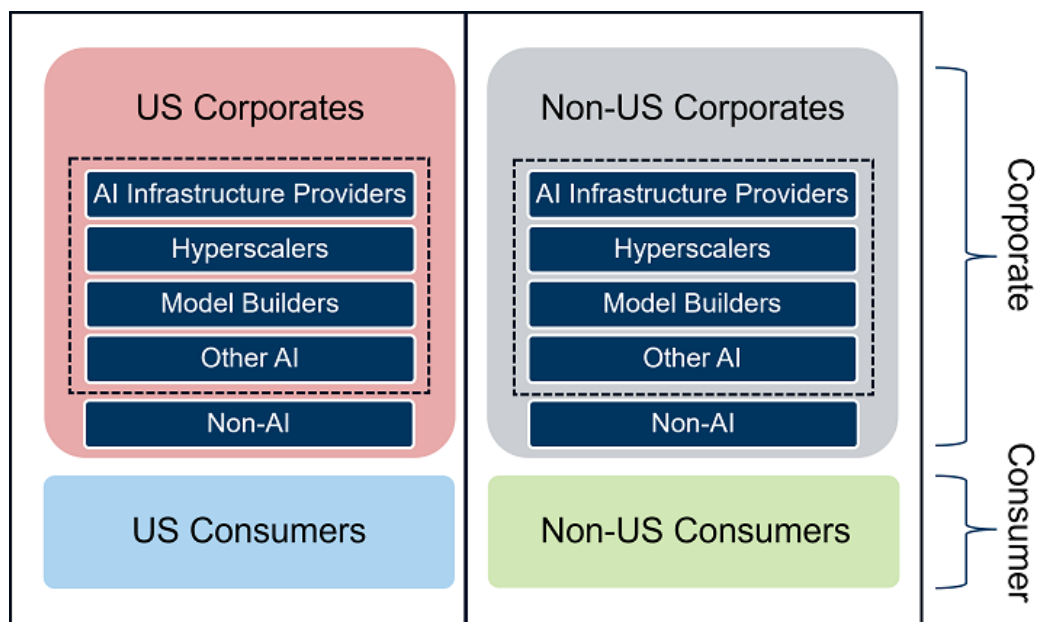
This set-up has, in our view, left the market vulnerable to any challenges to the more optimistic story. Robust earnings may dominate valuation concerns with the investment boom likely to continue to extend, but we still think investors should focus on the potential challenges to the theme and their consequences for portfolios. Our general view is that precision about the nature of any potential “shock” is key to understanding its cross-asset impact. The challenge is that there are a bewildering number of different possible narratives that could create risk to the current AI story. Although each may have its own specific characteristics, we think it useful to try to think of ways to organize and simplify these different kinds of worries.

Aggregate value or its distribution

In weighing those risks, we think a key distinction is whether a shock/risk shifts views on the likely *aggregate* value of AI-related innovation or on the *distribution* of that value (or both). Put simply, some worries are about the size of the pie, and some are about how the pie is shared. We have already shown that this distinction is critical to understanding recent trends in equity volatility, but it is also critical to the cross-asset footprint.

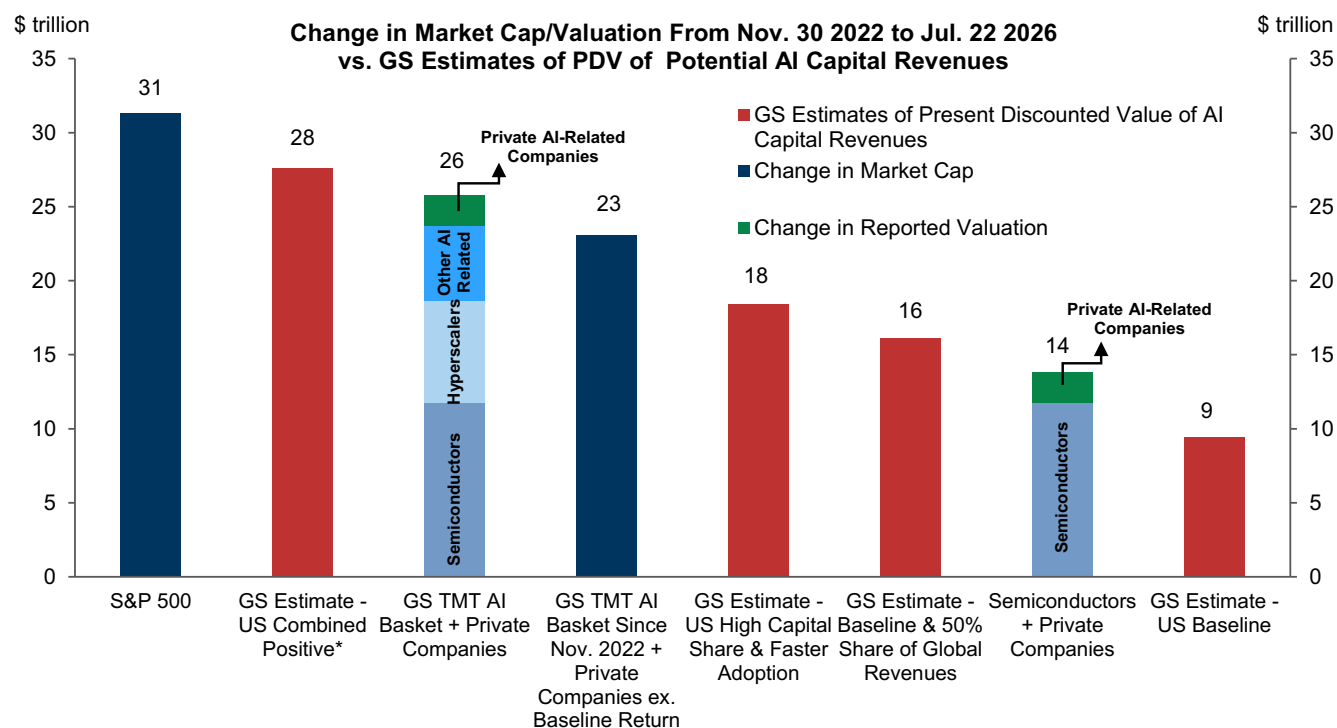
This distinction is embedded in the macro approach we have taken to looking at the value that we expect from AI. Our global economics team has calculated the Present Discounted Value of the additional GDP from the productivity gains that they expect from AI under our baseline assumptions. This defines the expected *aggregate* value of AI gains measured from today. To derive the value that should accrue to equity holders, we then need expectations of how that value is *distributed*. As Exhibit 1 shows, some portion of that should accrue to US companies in the form of higher profits. Those higher profits in turn will ultimately be shared among different types of companies: hyperscalers; infrastructure providers, like semiconductor, memory and power producers; model builders and other AI companies; and non-AI companies.¹ Some portion of those gains will also accrue to US consumers and workers and the same potential corporate and consumer beneficiaries are represented on the non-US side too.

¹ These “gains” could be negative for some segments if, for instance, AI-focused companies reduced profits for non-AI companies.

Exhibit 1: AI economic gains will ultimately be split across a wide range of groups

Source: Goldman Sachs Global Investment Research

So far, most of the AI-related value has accrued to US AI-focused companies and some key non-US companies, especially in Asia. The focus of our work so far has largely been on comparing estimates of the economic gains for US companies and the distribution of those gains across the major categories of US companies [Exhibit 2](#). From that narrower perspective, anything that distributes value away from US companies—away from the the top-left segment of [Exhibit 1](#)—including an increased share of AI gains going to consumers or non-US producers, is a shock to the “aggregate” value for the US equity market, even if the overall value of AI to the global economy does not change. We stick here to this narrower distinction between dynamics that affect the “aggregate” value of AI gains *for the US corporate sector* and those that affect its distribution across US companies. But it is easy to expand that view of what “aggregate” means through the lens of the framework that we set out here.

Exhibit 2: Our dominant focus has been on comparing value built into US companies and the potential capital revenues from AI


"Semiconductors", "hyperscalers", and "other AI-related" are constituents of the GS TMT AI basket (developed by GS Global Banking & Markets)
 *"US Combined Positive" assumes high capital share, faster adoption, and higher productivity gains from AI

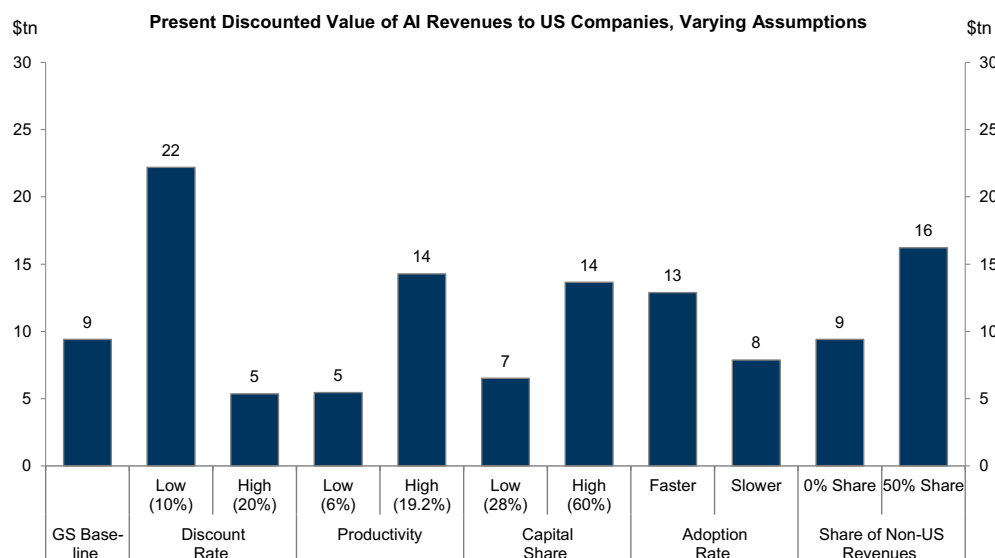
Source: Bloomberg, FactSet, Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Different drivers of aggregate and distributional shocks

The forces driving market views of the aggregate value of AI and its distribution are generally different. Perceptions of the aggregate value from AI are most likely to be affected by things that shift one or more of the five key parameters that underpin our global team's estimates of the value of future AI revenues to US companies (Exhibit 3):

- 1. Productivity growth:** the overall value of the innovation in improving work practices.
- 2. Adoption rate:** how quickly those productivity gains are being realized.
- 3. Capital share:** how much of those gains accrue to companies, which depends in turn on the ability to monetize the innovations.
- 4. International share:** how much can US companies capture a share of non-US gains or vice versa.
- 5. Discount rate:** the cost of capital for AI-related companies given financing needs and the uncertainty around potential revenues.

Exhibit 3: Views on the value of AI will shift if investors adjust their views on the key parameters



Source: Goldman Sachs Global Investment Research

Estimates of the PDV of AI revenues to US companies are sensitive to assumptions about each of these parameters, as we have illustrated. Macro shocks, which reduce the expected value for the entire corporate sector through assumptions about profits and the discount rate, even if the shock is not specifically related to AI, will also affect the aggregate value of US companies including AI-related companies.

For any given expectation of the aggregate value, the market can then make assumptions about who the winners and losers are likely to be, distributing that value across different parts of the corporate sector. Views on who will capture the benefits have shifted significantly over time. A prime example is the sharp increase in memory prices since late 2025, driven by shortages in the face of accelerating AI-related demand. That price increase is effectively a “terms of trade” shock that has helped producers and hurt consumers of chips. But beyond this dynamic, the market has also implicitly made judgments about the split between AI-related companies and the ultimate corporate users of AI products (assigning the bulk of AI-related value to the former) and about the capacity of AI disruption to shift profits away from incumbent firms in non-AI related industries (software, insurance, real estate). Shifts in views on issues that affect those divisions may not change expectations of aggregate profitability but will change how it is allocated.

Classifying AI concerns

We think using this framework to specify how a particular risk reduces or distributes value is a helpful exercise. Classifying different concerns along this axis can still be challenging (and open to challenge!). Some shocks are more obviously aggregate shocks; some are more obviously distributional; and for some it is less clear:

- Worries that focus on slower adoption or concern about the scope of use cases are likely to be *aggregate* shocks. They would lower estimates of some key parameters

that drive aggregate value (lower productivity growth, lower adoption rates).

- Difficulties in monetizing AI-related products or competition in models or applications that lower the cost of the innovation (as opposed to pessimism about the innovation itself) could function as a shock to the *aggregate* value that will accrue to US companies by distributing the gains more to consumers (lower capital share). But US corporate consumers could also benefit, and lower-cost products could accelerate adoption, which could shift the potential distribution of value towards them and mitigate the aggregate impact. Innovations by non-US companies could also *distribute* value away from US companies even if the broad value of AI-related innovations remains high.
- Reduced willingness from the market to finance the AI investment boom would also look more like an *aggregate* shock. But a decision to slow capital spending because infrastructure looks sufficient could have more of a *distributional* flavor, reallocating value to hyperscalers from suppliers of the investment boom.
- Capacity expansion or competition in the semis/memory space or model innovation to rely on fewer or cheaper chips could hurt semiconductor producers, but by alleviating shortages would benefit corporate consumers, so would potentially be a *distributional* impact. To the extent that those gains are passed through to final consumers too (instead of accruing to companies), it could also have some *aggregate* impact.
- Displacement or disruption of incumbent companies through fear of AI are obviously *distributional* at some level but could also have *aggregate* implications for US public market profits if they result in increased gains to consumers or if the corporate winners are outside the US public markets.
- Macro shocks—tighter monetary policy, a renewed rise in oil prices or a weakening in the labor market (AI-related or otherwise)—would function more like *aggregate* shocks, affecting value in the overall space. Given the value that has been built into AI-related companies, heavy positioning in many of these areas, and the financing needs of the AI boom, macro shocks could disproportionately affect AI beneficiaries vs. the rest of the broader market. But the nature of the macro shock likely matters here too, given that some shocks (higher oil prices, for instance) may affect other market segments disproportionately.

Different drivers, different asset footprints

The distinction between aggregate and distributional shock matters for two main reasons. It affects the cross-asset footprint; and it affects the volatility footprint.

For the cross-asset footprint, the main distinction is that aggregate shocks are likely to have much clearer impacts on the broad equity market and on a larger range of macro assets, especially interest rates, than distributional ones. We can illustrate this from the response to five market-moving events over the last 18 months.

1. **Deepseek (27 Jan 2025).** Release of DeepSeek's R1 model raised fears that US AI producers and their suppliers would capture less of the gains from AI.
2. **Broad AI worries (3 Feb–5 Feb 2026).** Hyperscaler earnings fueled worry about the

ability to sustain capital spending at high rates.

- 3. Payroll-driven rates spike (5 Jun 2026).** Strong payrolls release led to increased expectations of Fed tightening and higher yields across the US curve.
- 4. Google issuance (2 Jun 2026).** Google announces equity issuance to fund AI capex needs.
- 5. Meta cloud business (1 Jul 2026).** Meta announced that it would build out a new cloud business and sell excess compute.

Exhibit 4: Recent events illustrate the differing cross-asset footprint to different kind of shocks

	"Aggregate"/"Macro"			"Distributional"	
	Deepseek	Broad AI Worries	Payroll-Driven Rates Spike	Google Issuance	Meta Cloud Business
	Jan. 27 2025	Feb. 3 - Feb. 5 2026	Jun. 5 2026	Jun. 2 2026	Jul. 1 2026
Equities					
S&P 500	-1.5%	-1.7%	-2.6%	0.1%	-0.2%
Russell 2000	-1.0%	-2.7%	-3.5%	0.9%	-0.4%
Nasdaq 100	-3.0%	-3.1%	-4.8%	0.5%	-1.5%
Europe	-0.6%	-1.2%	-0.4%	1.4%	-0.6%
Japan	-3.5%	-0.4%	-5.5%	0.4%	-2.2%
Mainland China	0.9%	-1.3%	-2.0%	2.9%	1.2%
EM	-1.8%	-1.8%	-6.5%	1.0%	-2.8%
Korea	-2.4%	-3.3%	-14.1%	-1.0%	-8.1%
Taiwan	-5.6%	-1.4%	-7.3%	0.6%	-2.7%
VIX	20.5%	20.9%	39.7%	-1.7%	0.9%
GS US Cyclical (GSXUCYCL)	-0.2%	0.5%	-1.5%	1.0%	0.3%
GS US Defensives (GSXUDEFS)	1.9%	1.3%	1.1%	0.2%	0.5%
SPX ex. AI (SPXXAI)	1.2%	0.0%	0.0%	-0.2%	0.8%
GS US Broad AI Basket (GSTMATP)	-9.5%	-6.0%	-6.8%	3.2%	-4.5%
Semiconductors (SOXX)	-7.8%	-4.3%	-10.4%	5.8%	-6.4%
GS Power Up America (GSENEPOW)	-13.7%	-4.9%	-3.5%	2.6%	-3.6%
GS US Hyperscalers (GSXUHYPR)	-2.8%	-5.2%	-4.3%	-2.4%	2.5%
S&P 500 Software	-2.0%	-5.7%	-3.5%	-3.5%	3.1%
S&P 500 3m Avg. Implied Single Stock Volatility	4.1%	3.6%	-1.2%	0.5%	-1.2%
S&P 500 3m Implied Correlation	3bp	4bp	3bp	-0.1bp	0.4bp
S&P 500 3m Index Implied Volatility	11.1%	12.5%	8.6%	0.1%	0.3%
GS US Financial Conditions Index	7bp	10bp	20bp	-1bp	4bp
FX					
EUR/USD	0.0%	-0.2%	-0.8%	0.0%	-0.4%
AUD/USD	-0.5%	-0.7%	-1.3%	0.2%	-0.4%
JPY/USD	0.8%	-0.7%	-0.1%	-0.1%	0.1%
CAD/USD	-0.2%	-0.2%	-0.4%	0.1%	-0.1%
MXN/USD	-2.2%	-1.4%	-1.1%	0.4%	-0.3%
CNH/USD	-0.4%	-0.1%	0.0%	0.1%	-0.1%
TWD/USD	-0.5%	-0.3%	-0.4%	-0.4%	-0.2%
KRW/USD	-0.5%	-1.4%	-1.7%	-0.3%	-0.2%
ILS/USD	-1.0%	-1.5%	-1.1%	-1.1%	-0.4%
GS USD TWI	1.0%	0.5%	0.6%	0.0%	0.2%
Commodities					
BTC/USD	-3.6%	-17.1%	-3.1%	-5.4%	2.4%
Gold	-1.5%	-0.9%	-3.1%	0.3%	1.1%
Oil	-2.0%	0.1%	-2.7%	1.7%	-1.3%
Copper	-2.0%	-4.4%	-3.1%	1.6%	-0.7%
Rates					
UST 2y	-7bp	-12bp	10bp	1bp	0bp
UST 5y	-9bp	-11bp	8bp	0bp	1bp
UST 10y	-9bp	-9bp	6bp	-1bp	1bp
UST 30y	-8bp	-5bp	2bp	-1bp	2bp
Credit					
CDX IG	1bp	2bp	2bp	0bp	0bp
CDX HY	5bp	13bp	9bp	-1bp	0bp
Macro Shocks					
Growth shocks	-1.9	-2.2	-1.3	0.0	0.1
Policy shocks	-0.5	-0.2	2.8	-0.2	0.3

- Mainland China, EM, Korea, and Taiwan index returns are in USD terms

- FX changes are in XXX vs. USD terms. A positive change implies currency strength vs. USD, a negative change implies currency weakness vs. USD.

Source: Bloomberg, Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Exhibit 4 shows the performance of a range of assets across these episodes, while Exhibit 5 summarizes the asset market performance across several key assets (hyperscaler equities, semiconductor equities, SPX and US 10 year Treasury yield). These episodes are clearly not comprehensive but illustrate some of the possible differences in market reactions.

The first three episodes functioned as aggregate shocks.² The broad US equity market declined in each case, and credit spreads widened. Equities of both hyperscalers and companies supplying the investment boom (semiconductors, power suppliers) also fell, meaningfully underperforming the S&P 500. Yields fell in the first two episodes but rose in the episode where higher yields were the driver of the shock. Our standard macro toolkit show that asset prices moved in these three episodes consistent with significant macro shocks—resembling a growth downgrade for the first two and a hawkish policy shock in the third.

The final two episodes look much more like distributional shocks. On Google's issuance announcement, hyperscaler equities fell while companies supplying the investment boom rose, as the market determined the news as positive for the suppliers of the investment boom, but not for those driving the investment. With Meta's cloud announcement, the reverse was true. In both cases, the impact both on the broad US equity market and on US rate markets was limited. Large shifts under the surface of the equity market essentially netted out at the index level and shifts in our growth and policy shock measures were small.

The moves in other markets were less clear though the USD TWI rose in the three aggregate shocks (particularly against more cyclical currencies), while gold, bitcoin and copper fell. US defensive equities also rose in the aggregate shocks, despite declines at the index level.

Exhibit 5: Aggregate shocks have clear impacts on macro assets, while distributional ones do not

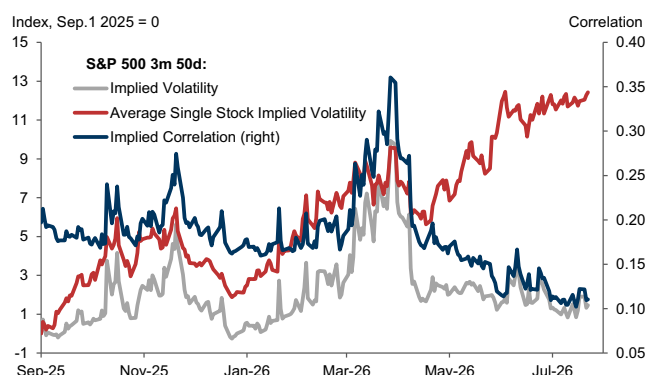
	"Aggregate"/"Macro"			"Distributional"	
	Deepseek	Broad AI Worries	Payroll-Driven Rates Spike	Google Issuance	Meta Cloud Business
Hyperscaler Equities	-	-	-	-	+
Semiconductor Equities	-	-	-	+	-
S&P 500	-	-	-	0	0
US 10y Yield	-	-	+	0	0

Source: Goldman Sachs Global Investment Research

We have argued before that the shifts in equity volatility over the last 6-9 months can also be seen through the distinction of aggregate ("macro") and distributional shocks. The rise in equity volatility and falls in implied correlations are an indication that so far it has been volatility around the distribution of AI gains that has dominated, except around key macro events like the onset of the Iran War ([Exhibit 6](#) and [Exhibit 7](#)). The episodes in [Exhibit 4](#) support the story that only aggregate/macro shocks have raised implied correlation and index volatility, supporting the idea that it is only worry about the aggregate value of AI gains that is likely to spill over into those broader measures.

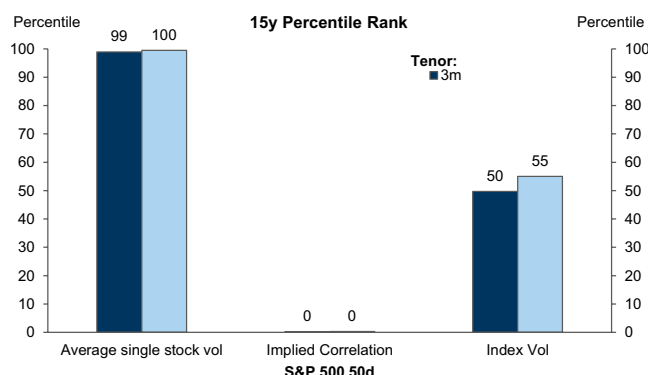
² As discussed earlier, innovations like Deepseek which potentially lower the cost of the innovation may redistribute value to consumers, including corporate consumers, but likely reduce perceptions of the overall share of AI gains going to the US corporate sector as a whole.

Exhibit 6: Single stock volatility has risen sharply over the last year while implied correlation has hit new lows



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Exhibit 7: High single stock volatility, low implied correlation reflects a focus on distributional issues



Source: Goldman Sachs FICC and Equities, Goldman Sachs Global Investment Research

Aggregate AI shocks more conducive to macro protection

Although the distinction we have drawn here between aggregate and distributional shocks is blunt, we think it has some important implications:

- The source of the shock to AI matters a lot to its cross-asset impact, so precision about the underlying drivers of the problem is essential. Slowing capex from the hyperscalers, for example, could come because of worries about the profitability of the innovations; from financing constraints; or from a realization that they are closer to what they need or have found more efficient ways to use existing investments than before. The impact of each version of that narrative will be different.
- The distinction between aggregate shocks and distributional shocks is central to predicting the impact on markets and volatility. Only aggregate/macro shocks are likely to have broad cross-market impact. So for any risk scenario, it is helpful to try to pin down whether it is one that is likely to reduce the aggregate value of AI gains or redistribute them. Even when it is hard to be definitive, we think it is a valuable exercise.
- For investors who are mostly dealing with broad-based US market exposures, macro assets are likely to be available as hedges. This is because distributional shocks are less likely to affect portfolio performance meaningfully. Diversification across countries and sectors also internalizes the volatility associated with shifts in views on who will win or lose from AI.
- For those with more specific AI exposures (or overweight positions), the task of protecting against shocks is more challenging. Protecting against aggregate and macro risks to the AI thesis remains possible, but exposure to shifts in the perceived winners and losers from AI are not as easily protected with other assets unless they are closely linked to the underlying positions. This is because many assets may move less reliably in relation to the core exposure, as distributional shock generally have offsetting effects at the macro level.
- For aggregate shocks, equity indices and rates appear to show clearer movements than FX and commodities. Except when higher rates are themselves the source of

the risk, worries about the aggregate gains from AI are likely to push US yields lower.

- Aggregate shocks should push implied correlation higher, while distributional shocks may not. This means that aggregate shocks have more scope to push index volatility up. The more distributional issues dominate, the more likely it is that pressure on index volatility will be muted. The impact on single stock volatility from either kind of shock may also be more limited given already high levels.

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Disclosure Appendix

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